RAVI SHANKAR
DAV – SECTOR 4

Exercise Set 6.1
Perimeter and Area

Unless stated otherwise, use $\frac{22}{7}$ for π.

Question 1

Question: The perimeter of a circle is 44 cm. What is its radius?

Solution

We know,
$C = 2\pi r$
Given,
$C = 44$ cm.
Therefore,
$44 = 2 \times \frac{22}{7} \times r$
$r = \frac{44 \times 7}{44}$
$\boxed{r = 7 \text{ cm}}$

Question 2

Question: Calculate, correct to 3 significant figures, the circumference of a circle with:

(i) radius 7 cm

(ii) radius 10 cm

(iii) radius 12 cm

Solution

We use,
$C = 2\pi r$
(i) When $r = 7$ cm
$C = 2 \times \frac{22}{7} \times 7$
$C = 44$ cm.
Therefore,
$\boxed{C = 44.0 \text{ cm}}$
(ii) When $r = 10$ cm
$C = 2 \times \frac{22}{7} \times 10$
$C = 62.857\ldots$
Therefore,
$\boxed{C = 62.9 \text{ cm}}$
(iii) When $r = 12$ cm
$C = 2 \times \frac{22}{7} \times 12$
$C = 75.428\ldots$
Therefore,
$\boxed{C = 75.4 \text{ cm}}$

Question 3

Question: Calculate the length of the arc of a circle if:

(i) the radius is 3.5 cm and the angle at the centre is 60°,

(ii) the radius is 6.3 cm and the angle at the centre is 120°.

Solution

Length of arc is
$s = \frac{\theta}{360^\circ} \times 2\pi r$
(i) $r = 3.5$ cm, $\theta = 60^\circ$
$s = \frac{60}{360} \times 2 \times \frac{22}{7} \times 3.5$
$s = \frac{1}{6} \times 22$
$\boxed{s = \frac{11}{3} \text{ cm}}$
(ii) $r = 6.3$ cm, $\theta = 120^\circ$
$s = \frac{120}{360} \times 2 \times \frac{22}{7} \times 6.3$
$s = \frac{1}{3} \times 39.6$
$\boxed{s = 13.2 \text{ cm}}$

Question 4

Question: Find the perimeter of a sector, i.e., the curved portion as well as the two straight portions, of a circle of radius 14 cm and sector angle 75°.

Solution

Perimeter of sector

$= 2r +$ length of arc

Arc length

$= \dfrac{75}{360} \times 2\pi \times 14$

$= \dfrac{5}{24} \times 28 \times \dfrac{22}{7}$

$= \dfrac{110}{12}$

$= \dfrac{55}{6}$ cm.

Therefore,

$P = 28 + \dfrac{55}{6}$

$P = \dfrac{223}{6}$

$\boxed{P \approx 37.17 \text{ cm}}$

Question 5

Question: Find the perimeters of the following shapes.

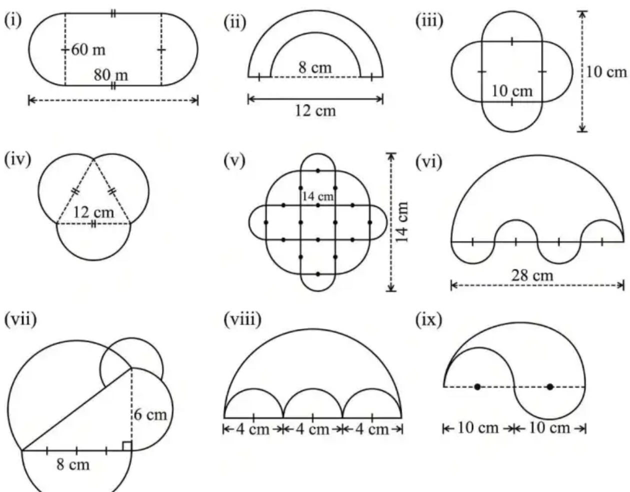

Fig. 6.14

Solution

(i)
There are two straight portions of 80 m each and two semicircles of diameter 60 m.
Together, two semicircles make one complete circle.
$P = 2(80) + \pi(60)$
$P = 160 + \frac{22}{7} \times 60$
$\boxed{P \approx 348.57 \text{ m}}$

(ii)
Outer semicircle has diameter 12 cm.
Inner semicircle has diameter 8 cm.
The two straight portions together measure
$12 - 8 = 4$ cm.
Therefore,
$P = \frac{\pi \times 12}{2} + \frac{\pi \times 8}{2} + 4$
$P = 6\pi + 4\pi + 4$
$P = 10\pi + 4$
$\boxed{P \approx 35.43 \text{ cm}}$

(iii)
The boundary consists of four semicircles, each of diameter 10 cm.
Four semicircles make two complete circles.
$P = 2\pi(10)$
$\boxed{P \approx 62.86 \text{ cm}}$

(iv)
There are three semicircles, each of diameter 12 cm.
$P = 3 \times \frac{\pi \times 12}{2}$
$P = 18\pi$
$\boxed{P \approx 56.57 \text{ cm}}$

(v)
There are four semicircles, each of diameter 14 cm.
$P = 4 \times \frac{\pi \times 14}{2}$
$P = 28\pi$
$\boxed{P = 88 \text{ cm}}$

(vi)
The large semicircle has diameter 28 cm.
The three smaller semicircles have diameter 7 cm each.
Therefore,
$P = \frac{\pi \times 28}{2} + 3\left(\frac{\pi \times 7}{2}\right)$
$P = 14\pi + \frac{21\pi}{2}$
$P = \frac{49\pi}{2}$
$\boxed{P = 77 \text{ cm}}$

(vii)
The three sides of the triangle are:
6 cm, 8 cm and 10 cm.
A semicircle is constructed on each side.
Therefore,

Question 6

Question: If the diameter of a car tyre is 56 cm, then:

(i) How far does the car need to travel for the tyre to complete one revolution?

(ii) How many revolutions does the tyre make if the car travels 10 km?

Solution

Diameter,
$d = 56$ cm.
(i) Distance in one revolution
Distance = circumference
$= \pi d$
$= \frac{22}{7} \times 56$
$\boxed{176 \text{ cm}}$

(ii) Number of revolutions
$10 \text{ km} = 10 \times 1000 \times 100$
$= 1,000,000$ cm.
Therefore,
Number of revolutions
$= \frac{1,000,000}{176}$
$\boxed{\approx 5681.82}$
So, the tyre makes approximately
$\boxed{5682 \text{ revolutions}}$.

Question 9

Question: In $\triangle ABC$, the midpoint of BC is D. Median AD is drawn. P is any point on AD. Show that
area($\triangle ABP$) = area($\triangle ACP$).

Solution

Since D is the midpoint of BC,
$BD = DC$.
Triangles ABP and ACP have the same altitude from B and C to AD.
Also,
$BD = DC$.
Therefore, their bases have equal corresponding heights.
Hence,
$\boxed{\text{area}(\triangle ABP) = \text{area}(\triangle ACP)}$

Question 10

Question: Given a square $ABCD$, let P be a point within it. Join PA, PB, PC, PD. What is the ratio of the areas of the red region and the green region?

Solution

Let the red regions be
$\triangle APB$ and $\triangle CPD$.
Let the green regions be
$\triangle BPC$ and $\triangle DPA$.
In a square, the sum of the areas of two opposite triangles is equal to the sum of the other two opposite triangles.
Therefore,
Area of red region = Area of green region.
Hence,
$\boxed{\text{Red : Green} = 1 : 1}$

Question 11

Question: In $\triangle ABC$, D is the midpoint of AB. P is any point on BC, and Q is a point on AB such that $CQ \parallel DP$. PQ is joined. Prove that

$$\text{Area}(\triangle BPQ) = \frac{1}{2}\text{Area}(\triangle ABC).$$

Solution

Let
$BC = x, \qquad AB = y, \qquad BP = p.$
Since D is the midpoint of AB,
$BD = \frac{y}{2}.$
Because
$CQ \parallel DP,$
$\triangle BQC$ and the corresponding triangles give
$\frac{BQ}{BD} = \frac{BC}{BP}.$
Thus,
$BQ = \frac{y}{2} \times \frac{x}{p}.$
So,
$BQ = \frac{xy}{2p}.$
Now,
$\text{Area}(\triangle BPQ) = \frac{1}{2} \times BP \times BQ$
$= \frac{1}{2} \times p \times \frac{xy}{2p}$
$= \frac{xy}{4}.$
Also,
$\text{Area}(\triangle ABC) = \frac{1}{2}xy.$
Therefore,
$\text{Area}(\triangle BPQ) = \frac{1}{2}\text{Area}(\triangle ABC).$
Hence proved.
$\boxed{\text{Area}(\triangle BPQ) = \frac{1}{2}\text{Area}(\triangle ABC)}$

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Exercise Set 6.2

Measuring Space: Perimeter and Area

Question 1

Question: Find the area of triangle ADE in Fig. 6.31.

Solution

From the figure,
$AD = 8$ cm
Height $= 10$ cm
We know,
$A = \frac{1}{2} \times \text{base} \times \text{height}$
$A = \frac{1}{2} \times 8 \times 10$
$\boxed{A = 40 \text{ cm}^2}$

Question 2

Question: The parallel sides of a trapezium are 40 cm and 20 cm. If its non-parallel sides are both equal, each being 26 cm, find the area of the trapezium.

Solution

Difference of parallel sides:
$40 - 20 = 20$ cm
Since the trapezium is isosceles,
$\frac{20}{2} = 10$ cm.
Using Pythagoras theorem,
$h^2 + 10^2 = 26^2$
$h^2 = 676 - 100 = 576$
$h = 24$ cm.
Now,
$A = \frac{1}{2}(a + b)h$
$= \frac{1}{2}(40 + 20)(24)$
$= 30 \times 24$
$\boxed{A = 720 \text{ cm}^2}$

Question 3

Question: Find the area of a triangle, given that its sides are 8 cm and 11 cm long, and its perimeter is 32 cm.

Solution

Third side:
$= 32 - (8 + 11)$
$= 13$ cm.
Thus, the three sides are
8 cm, 11 cm and 13 cm.
Semi-perimeter:
$s = \frac{32}{2} = 16$ cm.
Using Heron's formula,
$A = \sqrt{s(s-a)(s-b)(s-c)}$
$= \sqrt{16(16-8)(16-11)(16-13)}$
$= \sqrt{16 \times 8 \times 5 \times 3}$
$= \sqrt{1920}$
$\boxed{A = 8\sqrt{30}\ \text{cm}^2}$

Question 4

Question: The sides of a triangular plot are in the ratio 3 : 5 : 7; its perimeter is 300 m. Find its area.

Solution

Let the sides be
$3x, \quad 5x, \quad 7x.$
Since perimeter is 300 m,
$3x + 5x + 7x = 300$
$15x = 300$
$x = 20.$
Therefore, the sides are
60 m, 100 m and 140 m.
Semi-perimeter:
$s = \frac{300}{2} = 150$ m.
Using Heron's formula,
$A = \sqrt{s(s-a)(s-b)(s-c)}$
$= \sqrt{150(150-60)(150-100)(150-140)}$
$= \sqrt{150 \times 90 \times 50 \times 10}$
$\boxed{A = 1500\sqrt{3}\ \text{m}^2}$

Question 5

Question: One diagonal of a rhombus is twice as long as the other diagonal. If the rhombus has area 128 cm^2, find the length of the shorter diagonal.

Solution

Let the shorter diagonal be x cm.
Then the longer diagonal is $2x$ cm.
Area of rhombus:
$A = \frac{1}{2}d_1d_2$
Therefore,
$128 = \frac{1}{2} \times x \times 2x$
$128 = x^2$
$x = \sqrt{128}$
$\boxed{x = 8\sqrt{2}\text{ cm}}$
Hence, the shorter diagonal is
$\boxed{8\sqrt{2}\text{ cm}}$.

Question 6

Question: $ABCD$ is a parallelogram. P and Q are any two points on side AB. What can you say about the ratio

area($\triangle PCD$) : area($\triangle QCD$)?

Solution

Triangles PCD and QCD have the same base CD.
Also, P and Q lie on AB and
$AB \parallel CD$.
Therefore, both triangles have the same height.
Hence,
area($\triangle PCD$) = area($\triangle QCD$).
Therefore,
$\boxed{1 : 1}$

Question 7

Question: O is any point on the diagonal PR of a parallelogram $PQRS$. Prove that the areas of triangles PSO and PQO are equal.

Solution

Triangles PSO and PQO have the same base PO.
Also, S and Q are at equal perpendicular distances from diagonal PR.
Therefore, the two triangles have
same base and equal heights.
Hence,
$\boxed{\text{area}(\triangle PSO) = \text{area}(\triangle PQO)}$
Hence proved.

Question 8

Question: If the mid-points of the sides of a 4-gon are joined in order, prove that the area of the parallelogram thus formed will be half of the area of the given 4-gon.

Solution

Let $ABCD$ be the given 4-gon.
Let P, Q, R, S be the mid-points of
$AB,\ BC,\ CD,\ DA$ respectively.
Join P, Q, R, S.
Draw diagonal AC.
In $\triangle ABC$, P and Q are mid-points.
Therefore, by midpoint theorem,
$PQ \parallel AC$
and
$PQ = \frac{1}{2}AC.$
Similarly, in $\triangle ADC$,
$SR \parallel AC$
and
$SR = \frac{1}{2}AC.$
Thus,
$PQ \parallel SR$
and
$PQ = SR.$
Hence, $PQRS$ is a parallelogram.
The four corner triangles together occupy half of the area of $ABCD$.
Therefore, the remaining parallelogram occupies the other half.
Hence,

$$\boxed{\text{Area}(PQRS) = \frac{1}{2}\text{Area}(ABCD)}$$

Hence proved.

Question 9

Question:

In $\triangle ABC$, the midpoint of BC is D (Fig. 6.32). Median AD is drawn. P is any point on AD. Show that

area$(\triangle ABP) = $ area$(\triangle ACP)$.

Solution

Since D is the midpoint of BC,
$BD = DC$.
Now, in $\triangle ABD$ and $\triangle ACD$,
they have equal bases BD and DC and the same height.
Therefore,

$$\text{area}(\triangle ABD) = \text{area}(\triangle ACD) \tag{1}$$

Similarly, in $\triangle PBD$ and $\triangle PCD$,
$BD = DC$
and both triangles have the same height from P.
Therefore,

$$\text{area}(\triangle PBD) = \text{area}(\triangle PCD) \tag{2}$$

Subtracting (2) from (1),
$\text{area}(\triangle ABD) - \text{area}(\triangle PBD)$
$= \text{area}(\triangle ACD) - \text{area}(\triangle PCD)$
Hence,
$\boxed{\text{area}(\triangle ABP) = \text{area}(\triangle ACP)}$
Hence proved.

Question 10

Question:

Given a square $ABCD$, let P be a point within it. Join PA, PB, PC, PD (Fig. 6.33).

What is the ratio of the areas of the red region ($\triangle PAB$ and $\triangle PCD$) and the green region ($\triangle PBC$ and $\triangle PDA$)?

Solution

Let the side of the square be a.
Let the perpendicular distance of P from AB be h.
Then the distance of P from CD is
$a - h$.
Therefore,
$\text{area}(\triangle PAB) = \frac{1}{2}ah$
and
$\text{area}(\triangle PCD) = \frac{1}{2}a(a-h)$.
Thus, red area is
$\frac{1}{2}ah + \frac{1}{2}a(a-h)$
$= \frac{1}{2}a[h + a - h]$
$= \frac{a^2}{2}$.
Similarly, the distances of P from BC and AD add up to a.
Therefore,
$\text{Green area} = \frac{a^2}{2}$.
Hence,
Red area = Green area.
Therefore,
$\boxed{\text{Red : Green} = 1 : 1}$

Question 11

Question:

In $\triangle ABC$, D is the midpoint of AB. P is any point on BC, and Q is a point on AB such that $CQ \parallel DP$. PQ is joined (Fig. 6.34).

Prove that

$$\text{area}(\triangle BPQ) = \frac{1}{2}\text{area}(\triangle ABC).$$

Solution

Since D is the midpoint of AB,

$BD = DA$.

Therefore,

$\text{area}(\triangle BDP) = \text{area}(\triangle DAP)$

because they have equal bases BD and DA and the same height from P.

Now, since

$CQ \parallel DP$,

the corresponding triangles formed in the figure have equal heights.

Using the midpoint property, the area of $\triangle BPQ$ becomes half the area of $\triangle ABC$.

Therefore,

$$\boxed{\text{area}(\triangle BPQ) = \frac{1}{2}\text{area}(\triangle ABC)}$$

Hence proved.

Exercise Set 6.3

Area of a Circle and its Parts

Unless stated otherwise, use $\frac{22}{7}$ for π.

Question 1

Question:

Find the area of a sector of a circle with radius 7 cm if the angle of the sector is $60°$.

Solution

We know,
$A = \frac{\theta}{360°}\pi r^2$
Given,
$r = 7$ cm and $\theta = 60°$.
Therefore,
$A = \frac{60}{360} \times \frac{22}{7} \times 7^2$
$= \frac{1}{6} \times \frac{22}{7} \times 49$
$= \frac{77}{3}$
Hence,
$\boxed{A = \frac{77}{3} \text{ cm}^2}$
or
$\boxed{A = 25\frac{2}{3} \text{ cm}^2}$

Question 2

Question:
Find the area of a quadrant of a circle whose circumference is 44 cm.

Solution

Given,
$C = 44$ cm.
We know,
$C = 2\pi r$
Therefore,
$44 = 2 \times \frac{22}{7} \times r$
$r = 7$ cm.
A quadrant is $\frac{1}{4}$ of a circle.
Therefore,
$A = \frac{1}{4}\pi r^2$
$= \frac{1}{4} \times \frac{22}{7} \times 7^2$
$= \frac{77}{2}$
Hence,
$\boxed{A = 38.5 \text{ cm}^2}$

Question 3

Question:
The length of the minute hand of a clock is 7 cm. Find the area swept by the minute hand in 10 minutes.

Solution

The minute hand completes
$360°$ in 60 minutes.
Therefore, in 10 minutes,

$$\theta = \frac{10}{60} \times 360°$$

$$\theta = 60°.$$

Here,
$r = 7$ cm.
Area swept is the area of a sector.

$$A = \frac{\theta}{360°}\pi r^2$$

$$= \frac{60}{360} \times \frac{22}{7} \times 7^2$$

$$= \frac{77}{3}$$

Hence,

$$\boxed{A = \frac{77}{3}\ \mathrm{cm}^2}$$

Question 4

Question:

A chord of a circle of radius 10 cm subtends $90°$ at the centre. Find the area of the corresponding:

(i) minor sector,

(ii) major sector.

Use $\pi \approx 3.14$.

Solution

Given,
$r = 10$ cm.

(i) Minor sector

Here,
$\theta = 90°$.
Therefore,

$$A = \frac{90}{360} \times 3.14 \times 10^2$$
$$= \frac{1}{4} \times 3.14 \times 100$$
$$= 78.5.$$

Hence,

$$\boxed{A_{\text{minor}} = 78.5 \text{ cm}^2}$$

(ii) Major sector

Major angle
$= 360° - 90°$
$= 270°$.
Therefore,

$$A = \frac{270}{360} \times 3.14 \times 100$$
$$= \frac{3}{4} \times 314$$
$$= 235.5.$$

Hence,

$$\boxed{A_{\text{major}} = 235.5 \text{ cm}^2}$$

Question 5

Question:

A chord of a circle of radius 15 cm subtends an angle of 60° at the centre of the circle. Find the areas of the corresponding minor and major segments of the circle.

Use $\pi \approx 3.14$ *and* $\sqrt{3} \approx 1.73$.

Solution

Given,
$r = 15$ cm and $\theta = 60°$.

Step 1: Area of minor sector

$A_{\text{sector}} = \frac{60}{360} \times 3.14 \times 15^2$
$= \frac{1}{6} \times 3.14 \times 225$
$= 117.75 \text{ cm}^2$.

Step 2: Area of triangle

The triangle formed is equilateral with side 15 cm.
Area of equilateral triangle
$= \frac{\sqrt{3}}{4}a^2$
$= \frac{1.73}{4} \times 15^2$
$= 97.3125 \text{ cm}^2$.

Step 3: Minor segment

Minor segment
$=$ Minor sector $-$ Triangle
$= 117.75 - 97.3125$
$= 20.4375$.
Therefore,
$\boxed{A_{\text{minor segment}} \approx 20.44 \text{ cm}^2}$

Step 4: Major segment

Area of circle
$= 3.14 \times 15^2$
$= 706.5 \text{ cm}^2$.
Major segment
$=$ Circle $-$ Minor segment
$= 706.5 - 20.4375$
$= 686.0625$.
Therefore,
$\boxed{A_{\text{major segment}} \approx 686.06 \text{ cm}^2}$

Question 6

Question:

A car has two wipers which do not overlap. Each wiper has a blade of length 28 cm and sweeps through an angle of 120°. Find the total area cleaned at each sweep of the blades.

Solution

For each wiper,
$r = 28$ cm
and
$\theta = 120°$.
Area swept by one wiper:
$$A = \frac{120}{360} \times \frac{22}{7} \times 28^2$$
$$= \frac{1}{3} \times \frac{22}{7} \times 784$$
$$= \frac{2464}{3}$$
$$= 821\frac{1}{3} \text{ cm}^2.$$
There are two wipers.
Therefore,
Total area
$$= 2 \times 821\frac{1}{3}$$
$$= 1642\frac{2}{3}.$$
Hence,
$$\boxed{A = 1642\frac{2}{3} \text{ cm}^2}$$

Question 7

Question:

A chord of a circle of radius r subtends an angle of $60°$ at the centre of the circle.
Show that the area of the corresponding minor segment is
$$\pi r^2 \left(\frac{1}{6} - \frac{\sqrt{3}}{4\pi}\right).$$

Solution

Area of minor segment
= Area of minor sector − Area of triangle.

For 60° sector,
$A_{\text{sector}} = \frac{60}{360}\pi r^2$
$= \frac{\pi r^2}{6}.$

The triangle formed by the two radii is equilateral.
Therefore,
$A_{\text{triangle}} = \frac{\sqrt{3}}{4}r^2.$
Hence,
$A_{\text{segment}} = \frac{\pi r^2}{6} - \frac{\sqrt{3}}{4}r^2$
Taking πr^2 common,
$A_{\text{segment}} = \pi r^2\left(\frac{1}{6} - \frac{\sqrt{3}}{4\pi}\right).$
Hence proved.

$$\boxed{A = \pi r^2\left(\frac{1}{6} - \frac{\sqrt{3}}{4\pi}\right)}$$

Question 8

Question:

An equilateral triangle is inscribed in a circle of radius r. Show that the ratio of the area of the triangle to the area of the circle is

$$\frac{3\sqrt{3}}{4\pi} \approx 0.413.$$

Solution

For an equilateral triangle inscribed in a circle,
the side of the triangle is
$a = \sqrt{3}r$.
Therefore,
Area of triangle
$= \frac{\sqrt{3}}{4}a^2$
$= \frac{\sqrt{3}}{4}(\sqrt{3}r)^2$
$= \frac{3\sqrt{3}}{4}r^2$.
Area of circle
$= \pi r^2$.
Therefore,
Ratio
$= \frac{\frac{3\sqrt{3}}{4}r^2}{\pi r^2}$
$= \frac{3\sqrt{3}}{4\pi}$.
Using $\pi \approx 3.14$ and $\sqrt{3} \approx 1.73$,
Ratio
≈ 0.413.
Hence,
$$\boxed{\frac{\text{Area of triangle}}{\text{Area of circle}} = \frac{3\sqrt{3}}{4\pi} \approx 0.413}$$

Question 9

Question:

A square is inscribed in a circle of radius r. Show that the ratio of the area of the square to the area of the circle is
$\frac{2}{\pi} \approx 0.637$.

Solution

The diagonal of the square is the diameter of the circle.
Therefore,
$d = 2r$.
If the side of the square is a, then
$a\sqrt{2} = 2r$.
Thus,
$a = \sqrt{2}r$.
Area of square
$= a^2$
$= 2r^2$.
Area of circle
$= \pi r^2$.
Therefore,
Ratio
$= \frac{2r^2}{\pi r^2}$
$= \frac{2}{\pi}$.
Using $\pi \approx 3.14$,
$\frac{2}{3.14} \approx 0.637$.
Hence,
$$\boxed{\frac{\text{Area of square}}{\text{Area of circle}} = \frac{2}{\pi} \approx 0.637}$$

Question 10

Question:

A hexagon is inscribed in a circle of radius r. Show that the ratio of the area of the hexagon to the area of the circle is

$\frac{3\sqrt{3}}{2\pi} \approx 0.827$.

Can you see why the answer is exactly twice the answer to Question 8?

Solution

A regular hexagon can be divided into 6 equilateral triangles.
Each triangle has side r.
Therefore,
Area of one triangle
$= \frac{\sqrt{3}}{4}r^2$.
Area of hexagon
$= 6 \times \frac{\sqrt{3}}{4}r^2$
$= \frac{3\sqrt{3}}{2}r^2$.
Area of circle
$= \pi r^2$.
Therefore,
Ratio
$= \frac{\frac{3\sqrt{3}}{2}r^2}{\pi r^2}$
$= \frac{3\sqrt{3}}{2\pi}$.
Using $\pi \approx 3.14$ and $\sqrt{3} \approx 1.73$,
Ratio
≈ 0.827.
Hence,

$$\boxed{\frac{\text{Area of hexagon}}{\text{Area of circle}} = \frac{3\sqrt{3}}{2\pi} \approx 0.827}$$

Now compare with Question 8:
$\frac{3\sqrt{3}}{2\pi} = 2 \times \frac{3\sqrt{3}}{4\pi}$.
Therefore, the answer is exactly twice the answer to Question 8.

Exercise 6.3 – Answers

Question	Answer
Q1	$\frac{77}{3}$ cm^2 $= 25\frac{2}{3}$ cm^2
Q2	38.5 cm^2
Q3	$\frac{77}{3}$ cm^2
Q4(i)	78.5 cm^2
Q4(ii)	235.5 cm^2
Q5(i)	20.44 cm^2 approximately
Q5(ii)	686.06 cm^2 approximately
Q6	$1642\frac{2}{3}$ cm^2
Q7	$\pi r^2 \left(\frac{1}{6} - \frac{\sqrt{3}}{4\pi}\right)$
Q8	$\frac{3\sqrt{3}}{4\pi} \approx 0.413$
Q9	$\frac{2}{\pi} \approx 0.637$
Q10	$\frac{3\sqrt{3}}{2\pi} \approx 0.827$

End-of-Chapter Exercises

Measuring Space: Perimeter and Area

Unless stated otherwise, use $\frac{22}{7}$ for π.

Question 1

Question:

Identities in algebra can sometimes be shown as area relationships. For example, the figure shown corresponds to

$(a + b)^2 = a^2 + 2ab + b^2$.

Draw figures corresponding to the identities

$(a + b)(a - b) = a^2 - b^2$

and

$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$.

Solution

First identity:
$(a+b)(a-b)=a^2-b^2$
Take a square of side a.
Remove a square of side b.
The remaining area is
a^2-b^2.
The remaining figure can be rearranged to form a rectangle of sides $(a+b)$ and $(a-b)$.
Therefore,
$(a+b)(a-b)=a^2-b^2$.
Hence proved geometrically.

Second identity:
$(a+b+c)^2$
Take a square of side $(a+b+c)$.
Divide each side into parts a, b and c.
The square is divided into:
$a^2,\ b^2,\ c^2$
and two each of
$ab,\ bc,\ ca$.
Therefore,
$(a+b+c)^2$
$=a^2+b^2+c^2+2ab+2bc+2ca$.
Hence proved geometrically.

Question 2

Question:

An isosceles triangle has perimeter 40 cm; the equal sides are 15 cm each. Find the area of the triangle.

Solution

Equal sides = 15 cm each.
Base
$= 40 - 15 - 15$
$= 10$ cm.
Height bisects the base.
Therefore, half base
$= 5$ cm.
By Pythagoras theorem,
$h^2 = 15^2 - 5^2$
$= 225 - 25$
$= 200.$
Thus,
$h = 10\sqrt{2}$ cm.
Area
$= \frac{1}{2} \times 10 \times 10\sqrt{2}$
$= 50\sqrt{2}.$
Hence,
$\boxed{50\sqrt{2}\ \text{cm}^2}$

Question 3

Question:

An isosceles triangle has base 10 cm, and its area is 60 cm^2. What are the lengths of the equal sides?

Solution

Base = 10 cm.
Height
$= \frac{2 \times 60}{10}$
$= 12$ cm.
Half of the base
$= 5$ cm.
Let each equal side be x.
By Pythagoras theorem,
$x^2 = 12^2 + 5^2$
$= 144 + 25$
$= 169.$
Therefore,
$x = 13$ cm.
Hence,
$\boxed{13\ \text{cm},\ 13\ \text{cm}}$

Question 4

Question:

The area of a right-angled triangle is 54 sq. cm. One of its legs has length 12 cm. Find its perimeter.

Solution

Let the other leg be x.
$\frac{1}{2} \times 12 \times x = 54$
$6x = 54$
$x = 9$ cm.
Hypotenuse
$= \sqrt{12^2 + 9^2}$
$= \sqrt{144 + 81}$
$= 15$ cm.
Therefore,
Perimeter
$= 12 + 9 + 15$
$= 36$ cm.
Hence,
$\boxed{36 \text{ cm}}$

Question 5

Question:
The sides of a triangle are in the ratio $2 : 3 : 4$, and its perimeter is 45 cm. Find its area.

Solution

Let the sides be
$2x,\ 3x,\ 4x$.
Then,
$2x + 3x + 4x = 45$
$9x = 45$
$x = 5$.
Therefore, sides are
$10,\ 15,\ 20$ cm.
Semi-perimeter,
$s = \frac{45}{2} = 22.5$ cm.
By Heron's formula,
$A = \sqrt{s(s-a)(s-b)(s-c)}$
$= \sqrt{22.5(12.5)(7.5)(2.5)}$
$= 18.75\sqrt{15}$.
Hence,
$\boxed{A = 18.75\sqrt{15} \text{ cm}^2}$
or approximately
$\boxed{72.62 \text{ cm}^2}$.

Question 6

Question:
The sides of a triangle have lengths 7 cm, 24 cm, 25 cm. Find the area of the triangle in two different ways.

Solution

Method 1: Pythagoras theorem
$7^2 + 24^2$
$= 49 + 576$
$= 625$
$= 25^2.$
Therefore, the triangle is right-angled.
Area
$= \frac{1}{2} \times 7 \times 24$
$= 84 \text{ cm}^2.$

Method 2: Heron's formula
$s = \frac{7 + 24 + 25}{2}$
$= 28$ cm.
Therefore,
$A = \sqrt{28(28 - 7)(28 - 24)(28 - 25)}$
$= \sqrt{28 \times 21 \times 4 \times 3}$
$= \sqrt{7056}$
$= 84 \text{ cm}^2.$
Hence,
$\boxed{84 \text{ cm}^2}$

Question 7

Question:

If the wheel of a bicycle has a diameter of 60 cm, find how far a cyclist will have travelled after the wheel has rotated 100 times.

Solution

Diameter = 60 cm.
Circumference
$= \pi d$
$= \frac{22}{7} \times 60$
$= \frac{1320}{7}$ cm.
For 100 rotations,
Distance
$= 100 \times \frac{1320}{7}$
$= \frac{132000}{7}$ cm.
$= 18857.14$ cm.
Therefore,
$= 188.57$ m approximately.
Hence,
$\boxed{188.57 \text{ m}}$

Question 8

Question:

Find the area of a quadrant of a circle whose circumference is 66 cm.

Solution

Circumference = 66 cm.
$2\pi r = 66$
$2 \times \frac{22}{7} \times r = 66$
$r = 10.5$ cm.
Area of quadrant
$= \frac{1}{4}\pi r^2$
$= \frac{1}{4} \times \frac{22}{7} \times 10.5^2$
$= 86.625$ cm^2.
Hence,
$\boxed{86.625 \text{ cm}^2}$

Question 9

Question:
The wheel of a car has an outer radius of 28 cm. Calculate how far the car travels after one complete turn of the wheel, and how many times the wheel turns during a journey of 1 km.

Solution

Radius = 28 cm.
Circumference
$= 2\pi r$
$= 2 \times \frac{22}{7} \times 28$
$= 176$ cm.
Therefore, in one complete turn,
$\boxed{176 \text{ cm}}$
distance is covered.
For 1 km,
$= 100000$ cm.
Number of turns
$= \frac{100000}{176}$
$= \frac{6250}{11}$
≈ 568.18.
Hence,
$\boxed{\text{About 568 turns}}$

Question 10

Question:
Two rectangles have the same area and the same perimeter. Does this mean that they are congruent to each other?

Solution

Let the sides of a rectangle be a and b.
Area $= ab$.
Perimeter $= 2(a+b)$.
Suppose two rectangles have the same area and perimeter.
Then,
$a + b =$ constant
and
$ab =$ constant.
Thus, a and b are the two roots of the same quadratic equation.
Therefore, the two rectangles have the same dimensions, possibly with their length and breadth interchanged.
Hence, they are congruent.
$\boxed{\text{Yes, they are congruent.}}$

Question 11

Question:

You know that the area of a parallelogram is base $\times$ height. Using this and the figure, show that the area of a trapezium is half the sum of the parallel sides multiplied by the height:
$\frac{1}{2}(a+b)h$.

Solution

Take two identical trapeziums.
Join them to form a parallelogram.
The base of the parallelogram is
$a + b$
and its height is h.
Therefore,
Area of parallelogram
$= (a+b)h$.
This parallelogram consists of two equal trapeziums.
Therefore,
Area of one trapezium
$= \frac{1}{2}(a+b)h$.
Hence,
$\boxed{\text{Area of trapezium} = \frac{1}{2}(a+b)h}$

Question 12

Question:

By dividing a trapezium into two triangles show that its area is half the sum of the parallel sides multiplied by the height.

Solution

Let the parallel sides be a and b and height be h.
Draw a diagonal.
The trapezium is divided into two triangles.
Area of first triangle
$= \frac{1}{2}ah.$
Area of second triangle
$= \frac{1}{2}bh.$
Therefore,
Area of trapezium
$= \frac{1}{2}ah + \frac{1}{2}bh$
$= \frac{1}{2}(a+b)h.$
Hence,
$\boxed{A = \frac{1}{2}(a+b)h}$

Question 13

Question:

Show how we can use two identical copies of a trapezium to make a parallelogram. How will this give us the formula for the area of a trapezium?

Solution

Take two identical trapeziums.
Rotate one trapezium and join it to the other.
They form a parallelogram.
The base of the parallelogram is
$a+b$
and height is h.
Therefore,
Area of parallelogram
$= (a+b)h.$
Since it contains two identical trapeziums,
Area of one trapezium
$= \frac{1}{2}(a+b)h.$
Hence,
$\boxed{A = \frac{1}{2}(a+b)h}$

Question 14

Question:

Show that the area of a kite is half the product of its diagonals. Show this:
(i) using algebra, and
(ii) using geometry.

Solution

Let the diagonals of the kite be d_1 and d_2.
(i) Using algebra
Let the first diagonal be divided into x and y.
Then,
$x + y = d_1$.
The kite consists of two triangles.
Therefore,
$A = \frac{1}{2}d_2x + \frac{1}{2}d_2y$
$= \frac{1}{2}d_2(x + y)$
$= \frac{1}{2}d_2d_1$.
Hence,
$\boxed{A = \frac{1}{2}d_1d_2}$.

(ii) Using geometry
Draw a rectangle having length d_1 and breadth d_2.
Its area is
d_1d_2.
The kite occupies exactly half of this rectangle.
Therefore,
$A = \frac{1}{2}d_1d_2$.
Hence proved.

Question 15

Question:

Three problems about fitting congruent shapes together.

(i) Rectangle $ABCD$ has sides a, b, and rectangle $PQRS$ has sides $2a, 2b$. Show that $PQRS$ has 4 times the area of $ABCD$. Does this mean that 4 copies of rectangle $ABCD$ will fit into rectangle $PQRS$?

Solution

Area of $ABCD$
$= ab$.
Area of $PQRS$
$= (2a)(2b)$
$= 4ab$.
Therefore,
$\boxed{\text{Area}(PQRS) = 4\ \text{Area}(ABCD)}$.
Yes, four copies fit exactly.

(ii) $\triangle ABC$ has sides a, b, c, and $\triangle PQR$ has sides $2a, 2b, 2c$. Show that PQR has 4 times the area of ABC.

Solution

Let the area of $\triangle ABC$ be A.
By Heron's formula,
$A = \sqrt{s(s-a)(s-b)(s-c)}$.
For $\triangle PQR$, every side is doubled.
Therefore,
$A' = \sqrt{2s(2s-2a)(2s-2b)(2s-2c)}$
$= \sqrt{16s(s-a)(s-b)(s-c)}$
$= 4A$.
Hence,
$\boxed{A' = 4A}$.
Yes, 4 copies can fit exactly.

(iii) $\triangle ABC$ has sides a, b, c, and $\triangle PQR$ has sides $3a, 3b, 3c$. Show that PQR has 9 times the area of ABC.

Solution

Every side is multiplied by 3.
Therefore, by Heron's formula,
$A' = \sqrt{3s(3s-3a)(3s-3b)(3s-3c)}$
$= \sqrt{81s(s-a)(s-b)(s-c)}$
$= 9A$.
Hence,
$\boxed{A' = 9A}$.
Yes, 9 copies can fit exactly.

Question 16

Question:

What fraction of the triangle is shaded?
What fraction of the square is shaded?

Solution

(i) Triangle
Let the area of the whole triangle be A.
From the equal-area divisions in the figure,
Area of one required part
$= \frac{A}{6}$
and the other required part
$= \frac{A}{3}$.
Therefore,
Shaded area
$= \frac{A}{6} + \frac{A}{3}$
$= \frac{A}{6} + \frac{2A}{6}$
$= \frac{A}{2}$.
Hence, required fraction
$= \frac{A/2}{A}$
$= \boxed{\frac{1}{2}}$.

(ii) Square
The construction divides the square into 25 equal small parts.
The shaded region contains 5 parts.
Therefore,
Required fraction
$= \frac{5}{25}$
$= \boxed{\frac{1}{5}}$.

Question 17

Question:
What fraction of the rectangle is covered by the circles?

Solution

Let the radius of each circle be r.
For Fig. 6.45, there are 3 circles.
Length of rectangle
$= 6r$.
Breadth
$= 2r$.
Area of rectangle
$= 6r \times 2r$
$= 12r^2$.
Area of 3 circles
$= 3\pi r^2$.
Therefore,
Fraction covered
$= \frac{3\pi r^2}{12r^2}$
$= \boxed{\frac{\pi}{4}}$.

For Fig. 6.46, there are 4 circles.
Length
$= 8r$.
Area of rectangle
$= 8r \times 2r$
$= 16r^2$.
Area of circles
$= 4\pi r^2$.
Therefore,
Fraction
$= \frac{4\pi r^2}{16r^2}$
$= \boxed{\frac{\pi}{4}}$.
Thus, both figures give the same fraction.

Question 18

Question:

Use the above to make a conjecture about the area occupied by circles fitted into a rectangle in the manner shown. Test your conjecture for particular cases: 10 circles, 20 circles, 50 circles. Then prove your conjecture.

Solution

Let there be n circles of radius r.
Total area of circles
$= n\pi r^2$.
Length of rectangle
$= 2nr$.
Breadth
$= 2r$.
Therefore,
Area of rectangle
$= 2nr \times 2r$
$= 4nr^2$.
Hence,
Fraction covered
$= \dfrac{n\pi r^2}{4nr^2}$
$= \boxed{\dfrac{\pi}{4}}$.

For 10 circles:
$\dfrac{10\pi r^2}{40r^2} = \dfrac{\pi}{4}$.
For 20 circles:
$\dfrac{20\pi r^2}{80r^2} = \dfrac{\pi}{4}$.
For 50 circles:
$\dfrac{50\pi r^2}{200r^2} = \dfrac{\pi}{4}$.
Therefore, the conjecture is true.
$\boxed{\text{Fraction occupied by circles} = \dfrac{\pi}{4}}$

Question 19

Question:

The figure shows nine identical rectangles fitted together to make a large rectangle whose area is 72 cm^2. Find the perimeter of each small rectangle.

Solution

Let the length and breadth of each small rectangle be a and b.
There are 9 rectangles.
Therefore,
$9ab = 72$
$ab = 8.$
From the figure,
$4a = 5b.$
Thus,
$b = \frac{4}{5}a.$
Substituting,
$a\left(\frac{4}{5}a\right) = 8.$
Therefore,
$a^2 = 10.$
Hence,
$a = \sqrt{10}$ cm.
Now,
$b = \frac{4}{5}\sqrt{10}$ cm.
Perimeter
$= 2(a+b)$
$= 2\left(\sqrt{10} + \frac{4}{5}\sqrt{10}\right)$
$= \frac{18}{5}\sqrt{10}$ cm.
Hence,
$\boxed{P = \frac{18\sqrt{10}}{5}\text{ cm}}$

Question 20

Question:

Show that the areas of the shaded blue triangle and the shaded red triangle are equal.

Find a way of cutting up the blue triangle into some number of pieces and rearranging the pieces to cover the red triangle.

Solution

The opposite side is divided into three equal parts.
Therefore, the base of the blue triangle and the base of the red triangle are equal.
Let each base be b.
Both triangles have the same vertex.
Therefore, they have the same perpendicular height h.
Area of blue triangle
$= \frac{1}{2}bh.$
Area of red triangle
$= \frac{1}{2}bh.$
Therefore,
$\boxed{\text{Area of blue triangle} = \text{Area of red triangle}}.$
For rearrangement, cut the blue triangle along the indicated lines and translate the resulting pieces to the corresponding positions of the red triangle. Since the corresponding pieces have equal areas, they cover the red triangle exactly.
Hence proved.

Question 21

Question:

The figure shows a quarter circle in a square. Its centre is at one vertex, and it passes through two adjacent vertices. There are two semicircles on two adjacent sides as diameters. They create the shaded regions A and B.

Show that A and B have equal area.

Solution

Let the side of the square be s.
The radius of the quarter circle is s.
Therefore,
Area of quarter circle
$= \frac{1}{4}\pi s^2.$
Each semicircle has diameter s.
Therefore, radius
$= \frac{s}{2}.$
Area of one semicircle
$= \frac{1}{2}\pi \left(\frac{s}{2}\right)^2$
$= \frac{1}{8}\pi s^2.$
Thus, area of two semicircles
$= \frac{1}{4}\pi s^2.$
This is exactly equal to the area of the quarter circle.
The common overlapping part is present on both sides.
Therefore, the remaining shaded regions must have equal areas.
Hence,
$\boxed{\text{Area}(A) = \text{Area}(B)}.$

Question 22

Question:

In Fig. 6.50, four semicircles have been drawn within the given square whose side is 2 units. The centres of these semicircles are the midpoints of the sides. They create a 4-petalled flower. Find the perimeter and the area of this flower.

Solution

Side of square = 2 units.
Therefore, diameter of each semicircle = 2 units.
Hence,
$r = 1$ unit.

Perimeter

The boundary consists of four semicircular arcs.
Therefore,
$P = 4\pi r$
$= 4\pi.$
Hence,
$\boxed{P = 4\pi \text{ units}}.$
Using $\pi = \frac{22}{7}$,
$P = \frac{88}{7}$ units.

Area

Area of square
$= 2^2$
$= 4$ sq. units.
The two opposite semicircles together have area
$= \pi r^2$
$= \pi.$
Therefore, the two remaining regions together have area
$= 4 - \pi.$
By symmetry, the other two regions together also have area
$= 4 - \pi.$
Thus, area of flower
$= 4 - 2(4 - \pi)$
$= 2\pi - 4.$
Hence,
$\boxed{A = 2\pi - 4 \text{ sq. units}}.$
Using $\pi = \frac{22}{7}$,
$A = \frac{16}{7}$ sq. units.

Question 23

Question:

In Fig. 6.51, two concentric circles have a common centre O. A chord BC of the larger circle touches the smaller circle at A. The length of BC is l.

Show that the area of the green region enclosed between the two circles is $\frac{1}{4}\pi l^2$.

Solution

Let the radius of the smaller circle be x.
Let the radius of the larger circle be y.
Since BC touches the smaller circle at A,
$OA \perp BC$.
Also, the perpendicular from the centre to a chord bisects the chord.
Therefore,
$AB = AC = \frac{l}{2}$.
In right-angled triangle OAB,
$OA^2 + AB^2 = OB^2$.
Therefore,
$x^2 + \left(\frac{l}{2}\right)^2 = y^2$.
Thus,
$y^2 - x^2 = \frac{l^2}{4}$.
Area of green region
$= \pi y^2 - \pi x^2$
$= \pi(y^2 - x^2)$
$= \pi \times \frac{l^2}{4}$.
Hence,
$\boxed{\text{Area of green region} = \frac{\pi l^2}{4}}$.
Hence proved.

End of Chapter 6
Measuring Space: Perimeter and Area

Question 24

Question:

In Fig. 6.52, semicircles have been drawn on all the sides of a right-angled triangle as shown. Show that

Area(A) + Area(B) = Area(C).

Solution

Let the sides of the right-angled triangle be a, b and c, where c is the hypotenuse.
By Pythagoras theorem,
$a^2 + b^2 = c^2$.
Area of semicircle on side a:
$A = \frac{1}{2}\pi\left(\frac{a}{2}\right)^2$
$= \frac{\pi a^2}{8}$.
Similarly,
$B = \frac{\pi b^2}{8}$
and
$C = \frac{\pi c^2}{8}$.
Therefore,
$A + B$
$= \frac{\pi a^2}{8} + \frac{\pi b^2}{8}$
$= \frac{\pi(a^2 + b^2)}{8}$
$= \frac{\pi c^2}{8}$
$= C$.
Hence,
$\boxed{A + B = C}$.
Hence proved.

Question 25

Question:

Fig. 6.53 shows two circles passing through each other's centres. Find the area of the region enclosed by the two circles in terms of the common radius r.

Solution

Let the common radius of the two circles be r.
The common region consists of two equal circular segments.
For each circle, the angle at the centre is 120°.
Area of sector:
$= \frac{120}{360}\pi r^2$
$= \frac{\pi r^2}{3}$.
The triangle formed by the two radii has all sides equal to r.
Therefore, it is an equilateral triangle.
Area of equilateral triangle:
$= \frac{\sqrt{3}}{4}r^2$.
Therefore, area of one segment is
$= \frac{\pi r^2}{3} - \frac{\sqrt{3}}{4}r^2$.
There are two equal segments.
Hence, required area
$= 2\left(\frac{\pi r^2}{3} - \frac{\sqrt{3}}{4}r^2\right)$
$= \frac{2\pi r^2}{3} - \frac{\sqrt{3}r^2}{2}$.
Therefore,

$$\boxed{\text{Area} = \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right) r^2}.$$

Question 26

Question:

In Fig. 6.54, we see three triangles within a rectangle. The areas of the triangles are A, B, and C, as marked. Show that the area of the rectangle is

$$\frac{2(A+C)(B+C)}{C}.$$

Solution

Let the width and height of the rectangle be w and h.
Let the common vertical side of triangles A and C be x and the common horizontal side of triangles B and C be y.
For triangle C,
$C = \frac{1}{2}xy$.
Now, triangles A and C together have base x and their total horizontal height is the width w.
Therefore,
$A + C = \frac{1}{2}wx$.
Similarly, triangles B and C together have base y and their total vertical height is the height h.
Therefore,
$B + C = \frac{1}{2}hy$.
Multiplying,
$(A + C)(B + C)$
$= \frac{1}{2}wx \times \frac{1}{2}hy$
$= \frac{1}{4}whxy$.
But,
$C = \frac{1}{2}xy$.
Therefore,
$xy = 2C$.
Hence,
$(A + C)(B + C)$
$= \frac{1}{4}wh(2C)$
$= \frac{1}{2}whC$.
Therefore,
$wh = \frac{2(A + C)(B + C)}{C}$.
But wh is the area of the rectangle.
Hence,
$\boxed{\text{Area of rectangle} = \frac{2(A + C)(B + C)}{C}}$.
Hence proved.